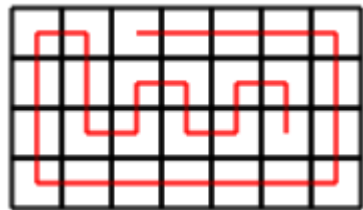


Grid Path Construction

Time limit: 1.00 s Memory limit: 512 MB

Given an $n \times m$ grid and two squares $a = (y_1, x_1)$ and $b = (y_2, x_2)$, create a path from a to b that visits each square exactly once.

For example, here is a path from $a = (1, 3)$ to $b = (3, 6)$ in a 4×7 grid:



Input

The first input line has an integer t : the number of tests.

After this, there are t lines that describe the tests. Each line has six integers n, m, y_1, x_1, y_2 and x_2 .

In all tests $1 \leq y_1, y_2 \leq n$ and $1 \leq x_1, x_2 \leq m$. In addition, $y_1 \neq y_2$ or $x_1 \neq x_2$.

Output

Print YES, if it is possible to construct a path, and NO otherwise.

If there is a path, also print its description which consists of characters U (up), D (down), L (left) and R (right). If there are several paths, you can print any of them.

Constraints

- $1 \leq t \leq 100$
- $1 \leq n \leq 50$
- $1 \leq m \leq 50$

Example

Input	Output
5	YES
1 3 1 1 1 3	RR
1 3 1 2 1 3	NO
2 2 1 1 2 2	NO
2 2 1 1 2 1	YES
4 7 1 3 3 6	RDL
	YES
	RRRRDDLLLLLLUUURDDRURDRURD